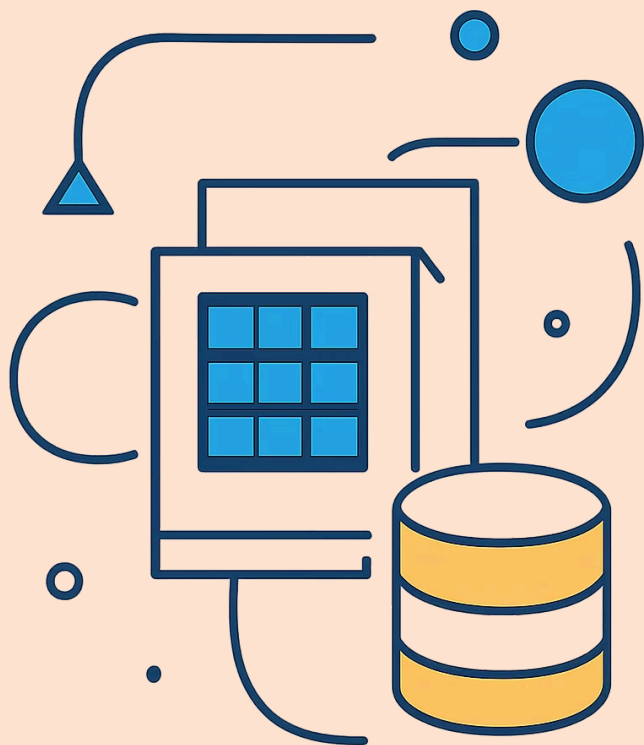


# Customer Shopping Behaviour Analysis

3,900 purchases | 18 columns | Goal: insights to guide strategy





# Dataset Summary

## Rows & Columns

3,900 rows · 18 columns

## Key Features

Demographics, Purchase details,  
Behavior, Subscription

## Missing Data

37 missing values in Review Rating

# Exploratory Data Analysis (Python)

## Load & Inspect

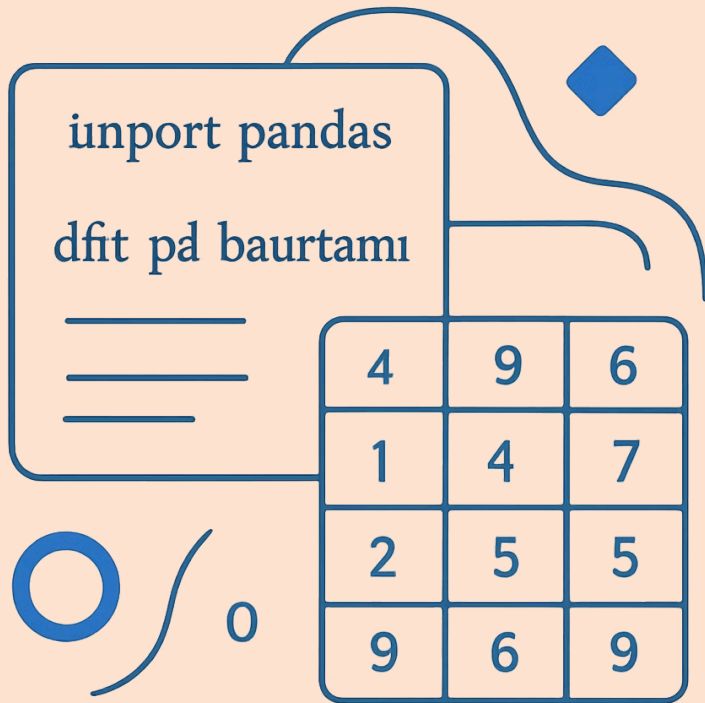
pandas: df.info(), .describe()

## Clean & Impute

Imputed Review Rating by category median

## Standardize

Renamed to snake\_case





## Feature Engineering & Prep

- Created age\_group bins
- Derived purchase\_frequency\_days
- Dropped promo\_code\_used (redundant with discount\_applied)
- Loaded cleaned data into PostgreSQL



# SQL Analysis — Key Business Questions

1

## Revenue by Gender

Female: 75,191 · Male: 157,890

2

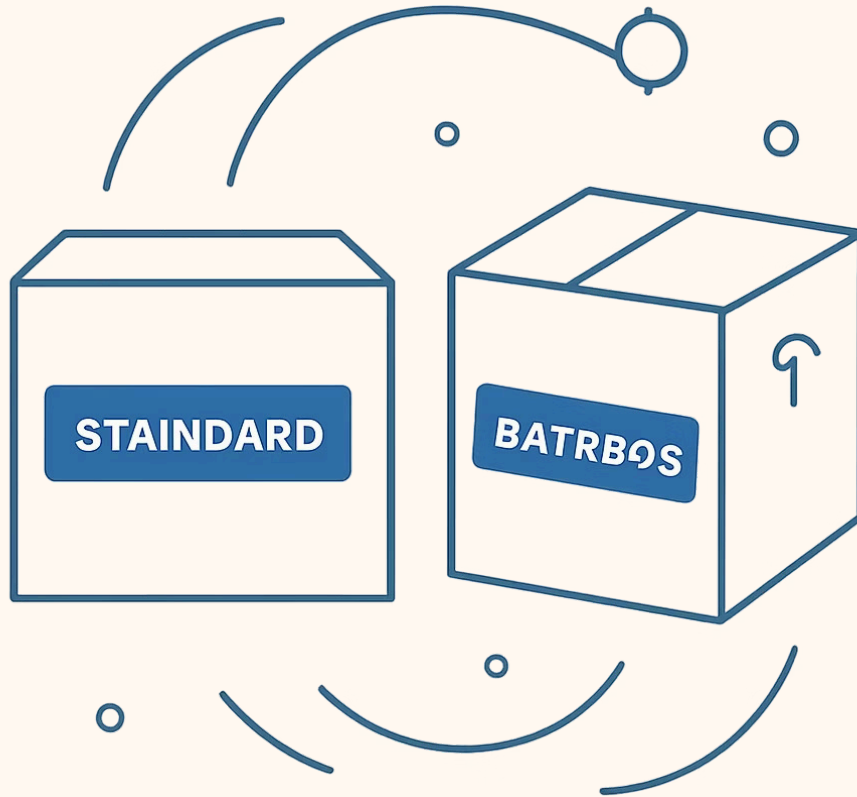
## High-Spending Discount Users

839 rows identified (discount used but above avg)

3

## Top Rated Products

Gloves, Sandals, Boots, Hat, Skirt (top 5 by avg rating)



## Shipping & Subscriptions

Avg purchase: Standard \$58.46 · Express \$60.48

Subscribers: 1,053 (avg \$59.49, total \$62,645)

Non-subscribers: 2,847 (avg \$59.87, total \$170,436)

# Discounts & Product Dependence



Hat

Discount rate 50%



Sneakers

49.66%



Coat

49.07%



Sweater

48.17%



Pants

47.37%

# Category Performance



## Clothing

Revenue ~100K · Sales ~1,800



## Accessories

Revenue ~60K · Sales ~1,200



## Footwear

Revenue ~30K · Sales ~600



## Outerwear

Revenue ~15K · Sales ~300





## Age & Segments

Revenue by age group:

- Young Adult ~62,143
- Middle-aged ~59,197
- Adult ~55,978
- Senior ~55,763

Customer segments: Loyal 3,116 · New 83 · Returning 701

# Business Recommendations

## Boost Subscriptions

Promote exclusive subscriber benefits

## Loyalty Programs

Reward repeat buyers to grow Loyal segment

## Review Discounts

Balance sales lift with margin control

## Product Positioning

Highlight top-rated & best-selling items

## Targeted Marketing

Focus on high-revenue age groups & express-shipping users

Power BI dashboard delivered for interactive exploration